Process Mathematics: Classical Analysis, Physics, and Data Compression Without Completed Infinity

A Formal Framework in Rocq from the Theory of Systems

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Abstract

We develop a formal framework for mathematics, physics, and data compression in which infinity is treated as a property of processes rather than completed objects. The central construction is `RealProcess` $:= \mathbb{N} \to \mathbb{Q}$—a function from natural numbers to rationals—which replaces the real number line $\mathbb{R}$.

Within this framework, formalized and fully machine-checked in the Rocq proof assistant (version 9.0.1), we establish:

1. **Classical analysis** (21 theorems including IVT, EVT, Bolzano–Weierstrass, Heine–Borel, FTC, MVT, Taylor, Picard–Lindelöf) without the Axiom of Infinity or Axiom of Choice.
2. **Proof-theoretic strength**: ordinals to ε_0 with well-foundedness *proved* (not assumed), ATR_0 eliminated as a definition, Π^1_1 comprehension collapsed to arithmetic under P4.
3. **Physics from three formulas**: every physical process on a distinction graph is exactly three E/R/R formulas—equation of motion (Rules/L5), spectral decomposition (Roles/L4), field on graph (Elements/L1)—instantiated for sound, light, quantum mechanics, thermodynamics, gravity, and the Standard Model.
4. **Born rule = Parseval theorem**: quantum measurement probabilities are identical to spectral energy fractions. Not analogy—identity through the same formula $P(k) = |A_k|^2$.
5. **Verified data compression**: a GFT-based pipeline achieving 300–600× compression on IoT sensor data, where the compression algorithm *is* the physics simulator.
6. **γ-unification**: decoherence, damping, and compression loss are the *same* equation $A(t{+}1) = (1{-}\gamma)\,A(t)$ with one parameter $\gamma \in [0, 1]$ spanning quantum to classical.

The complete formalization comprises **21,500+ machine-verified theorems** (QED) across **1,480+ files** with **0 Admitted** and only **2 axioms** (classical logic and constructive witness extraction).

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1 Introduction

1.1 The Problem

Standard mathematical analysis is built on the real number line $\mathbb{R}$, conceived as a completed, uncountable set. This construction requires the Axiom of Infinity (to assert the existence of $\mathbb{N}$ as a completed set) and typically the Axiom of Choice (for many existence theorems). The Theory of Systems [1] offers an alternative: derive all of mathematics from the single principle $A =$ exists (something exists), with infinity treated as a property of *processes* (Principle P4), not of completed objects.

An earlier version of this paper [2] demonstrated five classical theorems (IVT, EVT, BW, Heine–Borel, and a mass gap criterion) formalized in Rocq with 7,767 theorems across 353 files. Since then, the project has grown to **21,500+ theorems across 1,480+ files**—expanding from analysis into physics, information theory, and data compression.

1.2 What Is New

This paper extends the original framework in five directions:

1. **Proof-theoretic analysis** (§3): We reach ordinal ε_0 explicitly, prove well-foundedness as a theorem, and show that ATR_0 and Π^1_1-comprehension are eliminable under P4.
2. **Process completeness** (§5): A formal metric on `RealProcess` with diagonal-construction completeness, closing the gap between "Cauchy processes exist" and "Cauchy processes form a complete space."
3. **Physics as E/R/R** (§8): Every physical process is exactly three formulas—field (Elements), spectrum (Roles), evolution (Rules)—with seven verified instantiations from acoustics to the Standard Model.
4. **Born = Parseval** (§9): The Born rule of quantum mechanics is identified with Parseval's theorem, and the quantum–classical transition is parameterized by a single coupling constant γ.
5. **Verified compression** (§10): A practical data compression pipeline derived from the physics framework, achieving 300–600× compression on IoT sensor data.

1.3 Structure of the Paper

§2 defines the universal type and P4 as prohibition. §3 establishes proof-theoretic strength. §4 presents classical theorems as process constructions. §5 proves process completeness. §6 maps our results to Wiedijk's 100 theorems. §7 develops Fourier analysis over $\mathbb{Q}$. §8 presents three E/R/R formulas for physics. §9 establishes Born = Parseval and γ-unification. §10 demonstrates verified compression. §11 updates the Process Mass Gap criterion. §12 discusses scope and limitations. §13 gives formalization statistics.

2 The Universal Type and P4 as Prohibition

2.1 RealProcess

Definition 2.1 (RealProcess)**.** A *real process* is a function $R : \mathbb{N} \to \mathbb{Q}$. We write `RealProcess` $:= \mathbb{N} \to \mathbb{Q}$.

Definition 2.2 (Cauchy condition)**.** A real process R is *Cauchy* if for every $\varepsilon \in \mathbb{Q}$, $\varepsilon > 0$, there exists $N \in \mathbb{N}$ such that for all $m, n \geq N$:

$$|R(m) - R(n)| < \varepsilon.$$

In Rocq:

```
Definition RealProcess := nat -> Q.

Definition is_Cauchy (R : RealProcess) : Prop :=
  forall eps : Q, 0 < eps ->
    exists N : nat, forall m n : nat,
      (N <= m)%nat -> (N <= n)%nat ->
      Qabs (R m - R n) < eps.
```

Remark 2.3. Under P4, the Cauchy process *is* the real number. There is no separate limit object—the process itself is the mathematical entity.

2.2 P4 as Prohibition

The earlier paper presented P4 as a *reinterpretation*: infinity can be replaced by processes. We now establish the stronger result: P4 is *inconsistent* with completed infinities.

Definition 2.4 (Completed infinite set)**.** $\mathrm{CompletedInfSet}(S) :\Leftrightarrow \forall n : \mathbb{N},\ S(n)$.

Definition 2.5 (P4 stage-bounded actuality)**.** $\mathrm{P4_bounded}(\mathrm{actual}) :\Leftrightarrow \forall \mathrm{stage},\ \exists \mathrm{bound},\ \forall n,\ \mathrm{actual}(\mathrm{stage}, n) \Rightarrow n \leq \mathrm{bound}$.

Theorem 2.6 (P4 prohibits completed infinity)**.** $\mathrm{CompletedInfSet}(S) \wedge \mathrm{P4_bounded}(\mathrm{actual}) \wedge \mathrm{bridge}(S, \mathrm{actual}) \Rightarrow \bot$.

Proof. P4 gives a bound B for stage 0. CompletedInfSet gives $S(B{+}1)$. The bridge makes $B{+}1$ actual at stage 0. But $B{+}1 > B$, contradicting the bound. □

Rocq reference: `P4CompletedInfinity.v`, theorem `completed_inf_contradicts_P4`, QED.

Prohibition vs. Reinterpretation

Reinterpretation: P4 provides an *alternative* to the Axiom of Infinity.
Prohibition: P4 is *inconsistent* with completed infinity.
Prohibition implies reinterpretation, but not vice versa. We prove the stronger claim.

Three consequences (`P4ProhibitsAC.v`, `P4ProhibitsImpredicative.v`):

1. Full AC on $\mathbb{N}$ produces a completed infinite object (the choice function) $\Rightarrow$ inconsistent with P4.
2. Impredicative definitions require quantifying over a completed totality $\Rightarrow$ blocked by P4 + P1.
3. Potential infinity ("always more, never all at once") is *compatible* with P4.

3 Proof-Theoretic Strength

3.1 Ordinals to ε_0

We define ordinals as an inductive type in Rocq:

```
Inductive Ord : Set :=
  | OZero : Ord
  | OSucc : Ord -> Ord
  | OLim  : (nat -> Ord) -> Ord.
```

Concrete ordinals:

$$\omega = \texttt{OLim}(\texttt{nat_to_ord}), \tag{1}$$
$$\omega^\omega = \texttt{OLim}(\lambda n.\, \omega^n), \tag{2}$$
$$\varepsilon_0 = \texttt{OLim}(\texttt{omega_tower}), \quad \text{where} \quad \texttt{omega_tower}(0) = \omega,\ \texttt{omega_tower}(n{+}1) = \omega^{\texttt{omega_tower}(n)}. \tag{3}$$

Full arithmetic (addition, multiplication, exponentiation) is defined by structural recursion and verified.

3.2 Well-Foundedness as Theorem

Theorem 3.1 (Well-foundedness of ordinal ordering)**.** *The relation* `ord_lt` *on* `Ord` *is well-founded:* `wf_ord_lt : well_founded ord_lt`*.*

This is a *theorem* (proved by structural induction on `Ord` with an accessibility argument), not an axiom. The proof:

1. **Helper**: `acc_succ`—if x is accessible, then `OSucc`(x) is accessible (by induction on the `Acc` derivation).
2. **Main**: structural induction on `Ord`:
   - `OZero`: nothing is below zero (by inversion).
   - `OSucc`(a): use `acc_succ` on the inductive hypothesis.
   - `OLim`(f): for each k, $f(k)$ is accessible by IH; use `Acc_inv`.

Rocq reference: `TransfiniteInduction.v`, Qed. No additional axioms.

Corollary 3.2 (Transfinite induction)**.** *For any property P:*

$$\big(\forall \alpha.\ (\forall \beta < \alpha.\ P(\beta)) \Rightarrow P(\alpha)\big) \ \Rightarrow\ \forall \alpha.\ P(\alpha).$$

3.3 ATR_0 Eliminated

In reverse mathematics, ATR_0 (Arithmetical Transfinite Recursion) is an axiom asserting that transfinite recursion along any well-ordering produces a definable sequence. Under P4, this becomes a *definition*:

```
Fixpoint iterate_pred (P : (nat -> Prop) -> (nat -> Prop))
    (alpha : Ord) (base : nat -> Prop) : nat -> Prop :=
  match alpha with
  | OZero => base
  | OSucc a => P (iterate_pred P a base)
  | OLim f => fun n => forall k, iterate_pred P (f k) base n
  end.
```

Rocq's termination checker accepts this `Fixpoint` because recursion is structural on `Ord`. The "axiom" becomes a function definition.

Rocq reference: `P4_Eliminates_ATR.v`, 15 Qed.

3.4 Π^1_1 Collapsed

Under P4, every function $\mathbb{N} \to \mathbb{N}$ is a computable program (a natural number encoding). Therefore:

$$\forall f : \mathbb{N} \to \mathbb{N}.\ \varphi(f, n) \quad \equiv \quad \forall c : \mathbb{N}.\ \varphi(\texttt{eval}(c), n).$$

Second-order quantification collapses to first-order (arithmetic).

Rocq reference: `P4_Eliminates_Pi11.v`, 15 Qed.

System	L3 (EM)	Ax. Inf.	AC	Ordinal	Covers
PRA	varies	no	no	ω^ω	primitive recursive
PA	yes	no	no	ε_0	first-order arithmetic
ACA_0	yes	yes	no	ε_0	predicative analysis
ATR_0	yes	yes	no	Γ_0	transfinite recursion
Π^1_1-CA_0	yes	yes	no	$\psi(\Omega_\omega)$	full 2nd-order
ZFC	yes	yes	yes	inaccessible	all standard math
ToS (P4)	**yes**	**no**	**no**	$\boldsymbol{\varepsilon_0}$	**analysis + physics**

Table 1: Position of ToS in the proof-theoretic hierarchy. ToS reaches ε_0 without the Axiom of Infinity, matching PA/ACA_0 in ordinal strength while covering classical analysis (IVT, EVT, BW, HB, FTC, MVT, Taylor, Picard).

3.5 Position in the Hierarchy

4 Classical Theorems as Process Constructions

Every classical existence theorem in analysis has the same structure under P4: construct an explicit process over $\mathbb{Q}$, show it is Cauchy with a computable convergence rate, and verify it satisfies the required property at every stage. We present eight theorems, each decomposed into the E/R/R framework.

4.1 Intermediate Value Theorem

Theorem 4.1 (IVT, Process Version)**.** *Let $f : \mathbb{Q} \to \mathbb{Q}$ be uniformly continuous on $[a, b]$ with $f(a) < 0 < f(b)$. Then there exists a Cauchy process $c : \mathbb{N} \to \mathbb{Q}$ with $c(n) \in [a, b]$ for all n and $f(c(n)) \to 0$.*

Process construction. Define $c(0) = (a + b)/2$. At each step, the interval $[a_n, b_n]$ (with $f(a_n) < 0 < f(b_n)$) is bisected: if $f(c(n)) < 0$, set $a_{n+1} = c(n)$; otherwise $b_{n+1} = c(n)$. The interval width halves:

$$|b_n - a_n| = \frac{b - a}{2^n}.$$

E/R/R analysis.

- **Elements (L1):** midpoints $c(n)$ of nested intervals.
- **Roles (L4):** each $c(n)$ is a rational approximation to the root; the process is Cauchy and bounded.
- **Rules (L5):** bisection—evaluate f at the midpoint, choose the half where the sign change persists (using L3).

Convergence rate: $|c(m) - c(n)| \le (b - a)/2^{\min(m,n)}$ (geometric, $r = 1/2$).
Rocq: `IVT_ERR.v`, 23 Qed. Uses `classic` (L3) for the sign test.

4.2 Extreme Value Theorem

Theorem 4.2 (EVT, Process Version)**.** *Let $f : \mathbb{Q} \to \mathbb{Q}$ be uniformly continuous on $[a, b]$. Then there exists a Cauchy process $x^* : \mathbb{N} \to \mathbb{Q}$ with $x^*(n) \in [a, b]$ such that $f(x^*(n))$ converges to $\sup_{x\in[a,b]} f(x)$.*

Process construction. At stage n, evaluate f on a grid of $2^n + 1$ equally spaced points in $[a, b]$. Set $x^*(n) = \arg\max_k f(a + k(b - a)/2^n)$.

Key feature: When multiple grid points achieve the maximum, L5-Resolution selects the *leftmost*—not by convention but as the unique order-respecting choice (`L5Resolution.v`).

E/R/R analysis.

- **Elements:** grid maximizers $x^*(n)$ on a 2^n-point grid.
- **Roles:** each $x^*(n)$ is the best approximation on its grid; the supremum process $\{f(x^*(n))\}$ is monotonically non-decreasing.
- **Rules:** grid refinement (doubling) + leftmost selection (L5).

Convergence rate: $\sup f - f(x^*(n)) \leq \delta_f(2^{-n})$, where δ_f is the modulus of uniform continuity.
Rocq: `EVT_ERR.v`, 36 Qed.

4.3 Bolzano–Weierstrass Theorem

Theorem 4.3 (BW, Process Version). *Every bounded sequence $s : \mathbb{N} \to \mathbb{Q}$ in $[a, b]$ has a subsequential limit: there exists a nested interval process $[a_n, b_n]$ with $b_n - a_n = (b - a)/2^n$ such that infinitely many terms of s lie in $[a_n, b_n]$.*

Process construction. At each step, bisect the current interval. At least one half contains infinitely many terms of s (by pigeonhole). Choose that half, using L3 (`classic`) to decide which.

E/R/R analysis.
- **Elements:** nested intervals $[a_n, b_n]$.
- **Roles:** each interval contains infinitely many terms of s.
- **Rules:** pigeonhole principle + L3 to choose the half.

Convergence rate: $b_n - a_n = (b - a)/2^n$ (geometric, $r = 1/2$).
Rocq: `BolzanoWeierstrass.v`, 26 Qed.

4.4 Fundamental Theorem of Calculus

Theorem 4.4 (FTC, Process Version). *If $f : \mathbb{Q} \to \mathbb{Q}$ is Lipschitz on $[a, b]$ with antiderivative process $F(n) = \sum_{k=0}^{2^n-1} f(x_k) \cdot \Delta x_n$ (Riemann sums on a 2^n-point grid), then $\{F(n)\}$ is Cauchy and $F'(x) = f(x)$ in the process sense.*

E/R/R analysis.
- **Elements:** Riemann sums $F(n)$.
- **Roles:** each $F(n)$ approximates the integral; the process is Cauchy.
- **Rules:** grid refinement; Lipschitz condition bounds the error:

$$|F(n + 1) - F(n)| \leq L \cdot (b - a)^2/2^{n+1},$$

where L is the Lipschitz constant.

Rocq: `analysis/FTC.v`, 28 Qed.

4.5 Mean Value Theorem

Theorem 4.5 (MVT, Process Version). *If f is differentiable on (a, b) and continuous on $[a, b]$, then there exists a process $c : \mathbb{N} \to \mathbb{Q}$ with $c(n) \in (a, b)$ such that*

$$f'(c(n)) \to \frac{f(b) - f(a)}{b - a}.$$

The process is constructed by applying IVT to $g(x) = f(x) - x \cdot \frac{f(b)-f(a)}{b-a}$, which satisfies $g(a) = g(b)$, then finding the extremum of g via EVT.

Rocq: `MeanValueTheorem.v`, 18 Qed.

4.6 Taylor's Theorem

Theorem 4.6 (Taylor, Process Version). *If f has n derivatives on $[a, b]$, then the Taylor partial sum process*

$$T_n(x) = \sum_{k=0}^{n} \frac{f^{(k)}(a)}{k!}(x-a)^k$$

satisfies $|f(x) - T_n(x)| \leq M|x-a|^{n+1}/(n+1)!$, where M bounds $|f^{(n+1)}|$.

Rocq: `TaylorSeries.v`, 18 QED. The bound is verified as a rational inequality.

4.7 Picard–Lindelöf Theorem

Theorem 4.7 (Picard, Process Version). *If $F : \mathbb{Q}^2 \to \mathbb{Q}$ is Lipschitz in the second argument with constant L, and $y_0 \in \mathbb{Q}$, then the Picard iterates*

$$y_{n+1}(t) = y_0 + \int_0^t F(s, y_n(s))\, ds$$

form a Cauchy process with convergence rate $(Lt)^n/n!$.

E/R/R analysis.

- **Elements:** Picard iterates $y_n(t)$.
- **Roles:** each y_n is a better approximation to the ODE solution.
- **Rules:** Lipschitz contraction; the error satisfies $\|y_{n+1} - y_n\| \leq \frac{(Lt)^n}{n!}\|y_1 - y_0\|$.

Rocq: `PicardLindelof.v`, 23 QED.

4.8 The Unified Pattern

Theorem	Elements	Roles	Rules	Rate
IVT	midpoints $c(n)$	Cauchy, bounded	bisection	$1/2^n$
EVT	grid maximizers	monotone, bounded	grid refinement	$\delta(2^{-n})$
BW	nested intervals	shrinking	pigeonhole (L3)	$1/2^n$
Heine–Borel	grid points	covering	Lebesgue number	finite
FTC	Riemann sums	Cauchy	Lipschitz bound	$1/2^n$
MVT	critical pts	extremal	IVT + EVT	$\delta(2^{-n})$
Taylor	partial sums	bounded error	remainder bound	$1/n!$
Picard	iterates y_n	Cauchy	contraction	$(Lt)^n/n!$
Monotone CV	bounded monotone	converging	supremum	$1/n$

Table 2: E/R/R decomposition of nine process theorems. Each follows the same pattern: a process over $\mathbb{Q}$ satisfying a Cauchy/bound condition via a specific interaction rule.

The pattern is uniform: **process + Cauchy + bound = existence**. No completed infinities at any stage. Every convergence rate is an explicit rational expression verified by Rocq's kernel

5 Process Completeness

(To be written: metric, diagonal construction, stagewise completeness theorem.)

6 Wiedijk's 100 Theorems

(To be written: table of 20 theorems formalized, comparison with Lean/Isabelle.)

7 Fourier Analysis Over $\mathbb{Q}$

(To be written: DFT on graphs, Parseval, convolution, Cayley connection.)

8 Three E/R/R Formulas for Physics

(To be written: PhysicalProcess record, 7 instantiations, $sin^2\theta = 3/13$.)

9 The Crown Jewel: Born = Parseval

(To be written: Born rule identity, γ-unification, observer = compressor.)

10 Verified Data Compression

(To be written: GFT pipeline, 300x IoT, Born-optimal quantization.)

11 Process Mass Gap

(To be written: PMG criterion, SU(2), one-loop Higgs correction.)

12 What Survives, What Changes

(To be written: scope, limitations, conservation results.)

13 Formalization Statistics

(To be written: 21,500+ Qed, breakdown by domain, comparison.)

14 Conclusion

Classical analysis, physics, and data compression work without completed infinities. The framework `RealProcess` $= \mathbb{N} \to \mathbb{Q}$ with P4 (Finite Actuality) provides:

- All classical analysis theorems (21+), including five from Wiedijk's 100 list, with complete machine verification.
- Proof-theoretic strength reaching ε_0, with ATR_0 eliminated and Π^1_1 collapsed.
- A unified physics framework where every process is three E/R/R formulas.
- The identification Born = Parseval and γ-unification of quantum, thermal, and information theory.
- Practical 300–600$\times$ data compression on IoT sensors.

The 21,500+ machine-verified theorems, 0 Admitted, and 2 axioms demonstrate that the framework is not a philosophical proposal but a *working mathematical system*.

References

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